

Also Remember...

- If you are on a very tight budget and only need basic office productivity, you should go for the low-cost Cyrix MIII processor from VIA. This works with any socket-370 motherboard.
- If you play a lot of games you need to look for a processor that supports special instruction sets like 3DNow! or SSE.
- For high-end 3D design and CAD applications, you will require a processor with the highest clock speed you can afford and the largest amount of cache on board. This would make the Pentium and Athlon processors your best choice.

The Future of CPUs



Microprocessors have been around for over three decades. It was in 1971 that Intel introduced the 4004, its very first processor—which ran at 108 KHz to power the Busicom calculator. This started a revolution by making the device intelligent enough to carry mathematical functions. This processor was packed with total of 134,000 transistors.

It would be ridiculous to compare any modern day processor to the pioneering but lowly 4004. Current processors, such as the Intel Pentium 4 3.06 GHz Prescott, support 800 MHz FSB. These processors are designed around a dual execution unit architecture, delivering the extra performance you might get from two CPUs in a single package.

The future promises lots of excitement in the processor arena. Irrespective of demand CPUs continue to get faster, hitting not only 4 GHz but also the next big jump to in speed to 10 GHz. Pat Gelsinger, CEO of Intel, recently said that Intel's engineers are adding 25 MHz every week—and that this will come down to a day in a few years. While this is a tiny fraction of current clock speeds, its still an inexorable climb to more speed.

The 386SX processor in supported not more than 16 MB of memory, but the Itanium2 processor, which is waiting in the wings, will have as much as 6 MB of level 2 cache on it. The new processor will also consume lesser power. The lesser power requirement translates into lesser heat dissipation, which in turn means a longer life span. All this is made possible by developing new techniques that enable vendors to make smaller transistors that are so small that sub-atomic quantum effects become important; These techniques reduce width of wires that make up processor circuitry, lower current leakage by improving the fabrication process, and more.

As far as developing better performing processors goes, the common technique is to simply push for ever greater core clock and FSB speeds to get higher performance, while others approaches from companies such as IBM take the route of fabricating two processors on one chip to achieve the same goal of better performance. Workstation-level computing power continues its migration to desktops.

But as newer processors with newer architectures are released, the upgrade path becomes correspondingly more difficult than it used to

be. Newer processors will need mainboards that have chipsets specifically designed for them. A good example is the Intel 3.06 GHz processor, which supports 800 MHz FSB. This will not work at 800 MHz FSB on older boards, and clock down to 533 MHz. Hence one will have to buy a new mainboard to support the gigahertz monster.

64-bit Computing

After having tested the route to success by increasing the core clock of the process and FSB, what could be next on vendors' minds?

The answer: 64-bit architecture over the currently available 32-bit architecture.

Processors will soon embrace 64-bit architecture. The Opteron from AMD and the Itanium2 from

Intel are just two of the most-talked about 64-bit processors. These processors are currently available only for the server segment, but processors for the desktop should be available in a few years.

It will take some time for software vendors to come up will software optimised for these processors—a time-consuming task, but one that is inevitable. There are other vendors such as Sun, IBM and HP that are in the race to keep their customer base intact by offering 64-bit platforms of their own, or implementing Intel and AMD solutions in their architecture.

Embedded chips

The future will see embedded microprocessors grow unfettered. While embedded chips, which are inexpensive, are often treated as poor cousins, they still account for 98 per cent of total processors sold. Typically innovation developed for more expensive chips trickle down to embedded chips in a year or two. And some companies focus exclusively on developing embedded chips, which requires a special kind of hardware-software synergy.

In the end...

There's a whole world existing outside the realm of Intel and AMD. Processors from these vendors are used more in the handheld and mobile consumer products because of their lower power consumption and affordable price. But since such vendors are positioned in a completely different segment, they have very little presence in desktop computing, but will definitely be making a place in our homes and offices soon.

